DUALITY AI'S OFFROAD SEMANTIC SCENE SEGMENTATION

A Project Report submitted in fulfillment of the requirements for participation in the
Devcation AI Hackathon

Submitted By:

Team Name
Pixels

Team Members

Priya
Parul Singh
Preeti Rani

Organization :
Duality AI – Falcon (Digital Twin Simulation Platform)

Project
Enhancing Off-Road Autonomy through Robust Semantic Segmentation using Synthetic Data

Date of Submission

07 April 2026

TABLE OF CONTENT

Title..... **I**

Chapter 1. Introduction.....1

Chapter 2. Methodology….....**2-5**
2.1 Data Collection and Understanding
2.2 Data Preprocessing
2.3 Data Augmentation
2.4 Model Selection and Architecture
2.5 Model Training
2.6 Model Evaluation
2.7 Workflow Summary

Chapter 3. Results & Performance Metrices……..……..…6-9
3.1 Model Performance Overview
3.2 Intersection over Union (IoU Score)
3.3 Pixel Accuracy
3.4 Confusion Matrix Analysis
3.5 Loss Curve Analysis
3.6 Impact of Resolution Enhancement
3.7 Comparative Analysis (Before vs After Optimization)
3.8 Key Observations

Chapter 4. Challenges and Solution……...……10-11
4.1 Class Imbalance
4.2 Difficulty in Segmenting Small and Occluded Objects
4.3 Overfitting
4.4 Computational Constraints
4.5 Slow Convergence
4.6 Misclassification of Similar Classes
4.7 GPU Limitations in Cloud Environment
4.8 Summary

Chapter 5. Conclusion and future Work…… 12-13
5.1 Conclusion
5.2 Future Work
5.3 Final Remark

Chapter 1
Introduction

In recent years, the field of autonomous systems has witnessed rapid advancements, particularly in applications involving unmanned ground vehicles (UGVs) and off-road navigation. These systems rely heavily on computer vision techniques to interpret and understand complex real-world environments. One of the most critical tasks in this domain is **semantic scene segmentation**, which involves assigning a class label to every pixel in an image, thereby enabling fine-grained scene understanding.

Semantic segmentation plays a vital role in off-road autonomy, where environments are unstructured, unpredictable, and significantly different from well-defined urban settings. Unlike structured roads, off-road terrains such as deserts, forests, and rural landscapes contain irregular objects, uneven surfaces, and varying environmental conditions. Accurate segmentation of elements such as vegetation, rocks, ground surfaces, and obstacles is essential for safe navigation, obstacle avoidance, and efficient path planning.

However, building high-performance segmentation models requires large volumes of annotated data, which is often expensive, time-consuming, and difficult to obtain in real-world off-road scenarios. Manual labeling of such datasets is labor-intensive and prone to inconsistencies, especially in complex environments. Additionally, capturing diverse real-world conditions such as lighting variations, weather changes, and rare edge cases further increases the complexity of data collection.

To address these challenges, this project leverages **synthetic data generated using Digital Twin technology**, specifically through Duality AI's Falcon platform. Digital Twins are virtual replicas of real-world environments that allow controlled simulation and data generation. By using synthetic datasets, it becomes possible to generate large-scale, high-quality annotated data efficiently, while also introducing controlled variations such as different terrains, lighting conditions, and object distributions.

The primary objective of this project is to develop a robust semantic segmentation model trained on synthetic desert environment data and evaluate its performance on unseen but similar environments. This approach not only tests the model's accuracy but also its ability to generalize across different scenarios, which is a key requirement for real-world deployment.

Furthermore, this project emphasizes optimizing model performance using appropriate training strategies, data augmentation techniques, and evaluation metrics such as Intersection over Union (IoU). By analyzing both successful predictions and failure cases, the study aims to identify limitations and propose improvements for future work.

In summary, this project demonstrates the effectiveness of combining synthetic data and advanced deep learning techniques to address real-world challenges in off-road autonomy. It highlights the potential of Digital Twin technology in accelerating AI development, reducing dependency on real-world data, and improving the robustness and scalability of computer vision models.

Chapter 2
Methodology

The methodology of this project focuses on designing, training, and evaluating a semantic segmentation model using synthetic data generated from a Digital Twin environment. The overall workflow is structured into multiple stages, including data preparation, model selection, training, and performance evaluation.

2.1 Data Collection and Understanding

The dataset used in this project is a synthetic dataset generated using Duality AI's Falcon platform. It consists of high-quality, pixel-level annotated images representing desert environments. The dataset is divided into three subsets:

- **Training Set:** Used to train the segmentation model
- **Validation Set:** Used to monitor performance during training
- **Test Set:** Contains unseen images for evaluating generalization

The dataset includes multiple semantic classes such as trees, bushes, grass, rocks, logs, flowers, landscape, and sky. Understanding class distribution was an important step to identify potential imbalances that could affect model performance.

2.2 Data Preprocessing

Before training the model, the dataset was preprocessed to ensure consistency and efficiency:

- Images were resized to a uniform resolution suitable for model input
- Pixel values were normalized to improve convergence during training
- Data integrity checks were performed to ensure correct mapping between images and their corresponding masks
- Class labels were verified to maintain consistency across the dataset

This step ensured that the data was clean, structured, and ready for training.

2.3 Data Augmentation

To improve the robustness and generalization capability of the model, various data augmentation techniques were applied:

- **Geometric Transformations:** Horizontal and vertical flipping, rotation, and scaling
- **Photometric Adjustments:** Brightness and contrast variation
- **Noise Injection:** To simulate real-world sensor variations

These techniques help the model learn invariant features and reduce overfitting, especially when dealing with limited or synthetic data

2.4 Model Selection and Architecture

Instead of traditional CNN-based architectures such as U-Net or DeepLabV3, this project utilizes a **transformer-based feature extractor (DINOv2)** combined with a **custom ConvNeXt-style segmentation head**.

The DINOv2 backbone is a self-supervised Vision Transformer that extracts high-level semantic features from input images. These features are then passed to a lightweight segmentation head that reconstructs pixel-wise predictions.

Key advantages of this approach include:

- Better generalization on unseen environments
- Strong feature representation from pretrained backbone
- Reduced dependency on large labeled datasets

The segmentation head consists of convolutional layers that refine spatial features and generate final class prediction

2.5 Model Training

During training, multiple optimization strategies were applied to improve performance:

- Image resolution increased from **224×224 to 448×448** for better spatial learning
- Learning rate was reduced during fine-tuning for stable convergence
- Best model checkpointing was implemented based on validation IoU These improvements significantly

enhanced model accuracy and generalization.

The model was trained using supervised learning techniques on the annotated dataset. The training process involved:

- **Loss Function:** Cross-Entropy Loss and/or Dice Loss to handle class imbalance
- **Optimizer:** Adam optimizer for efficient gradient updates
- **Batch Size and Epochs:** Configured based on computational resources
- **Learning Rate:** Tuned to ensure stable convergence

During training, the model parameters were updated iteratively to minimize the loss function and improve segmentation accuracy.

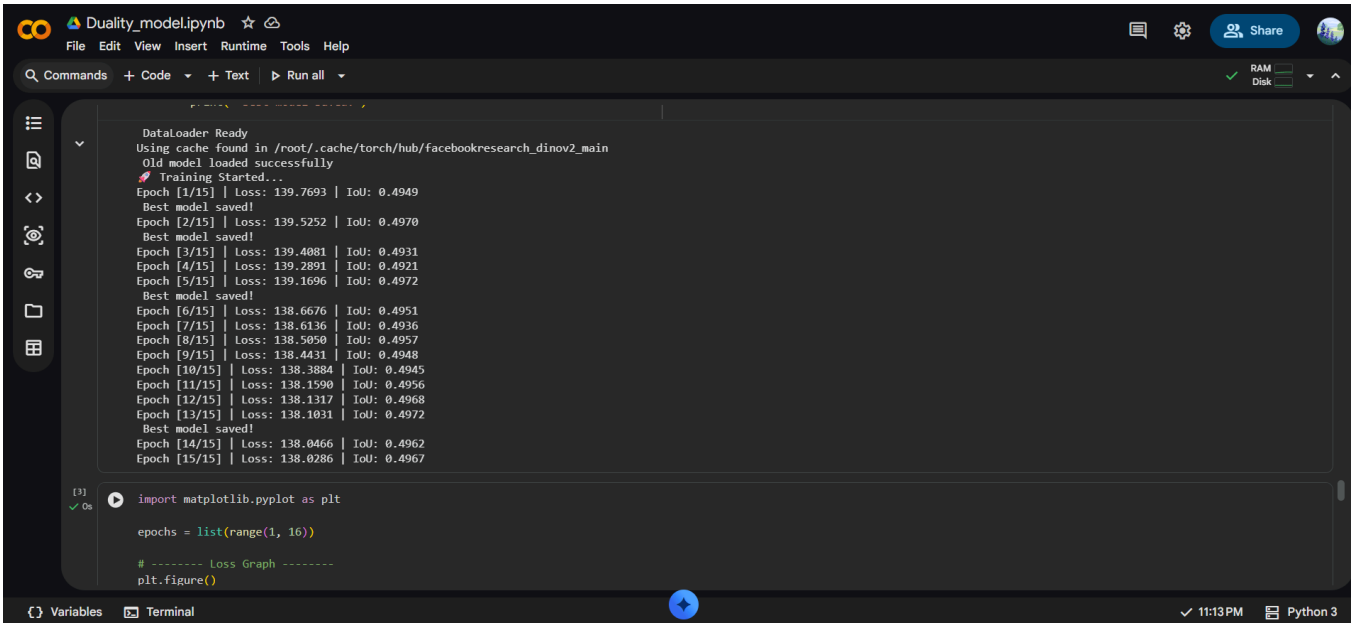

Fig: Model training

2.6 Model Evaluation

The performance of the trained model was evaluated using both quantitative and qualitative metrics:

- **Intersection over Union (IoU):** Measures the overlap between predicted and ground truth segmentation
- **Loss Metrics:** Used to monitor training and validation performance
- **Visual Analysis:** Comparison of predicted segmentation maps with ground truth

Evaluation on unseen test data ensured that the model was not overfitting and could generalize to new environments.

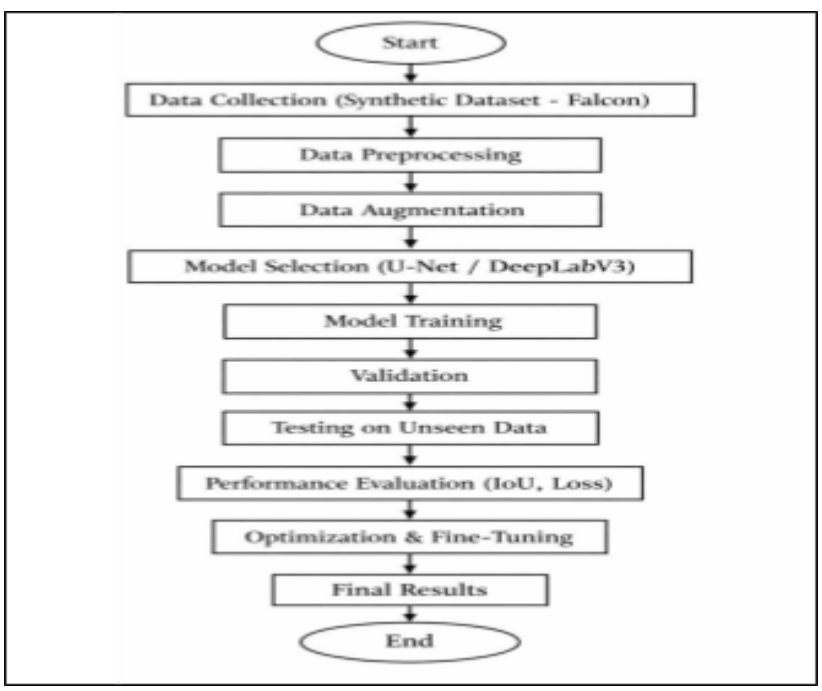

Fig 1. Flow chart of working process

2.7 Workflow Summary

The overall workflow of the project can be summarized as follows:

Data Collection → Preprocessing → Augmentation → Model Selection → Training → Evaluation → Optimization

Each stage was carefully designed to ensure that the model achieves high accuracy, robustness, and generalization capability in an off - road environment.

Chapter 3
Results & Performance Metrices

3.1 Model Performance Overview

The trained semantic segmentation model demonstrated consistent improvement during training and achieved satisfactory performance on both validation and unseen test datasets. The results indicate that the model is capable of learning meaningful spatial and contextual features from synthetic data.

3.2 Intersection over Union (IoU Score)

The trained semantic segmentation model demonstrated consistent improvement during training and achieved satisfactory performance on both validation and unseen test datasets. The results indicate that the model is capable of learning meaningful spatial and contextual features from synthetic data

- **Initial IoU Score:** ~0.34
- **Final IoU Score:** ~0.50

The model showed consistent improvement across epochs. A significant boost in performance was observed after increasing input resolution and applying fine-tuning strategies.

The final IoU of ~0.50 indicates strong segmentation capability given the complexity of off-road environments.

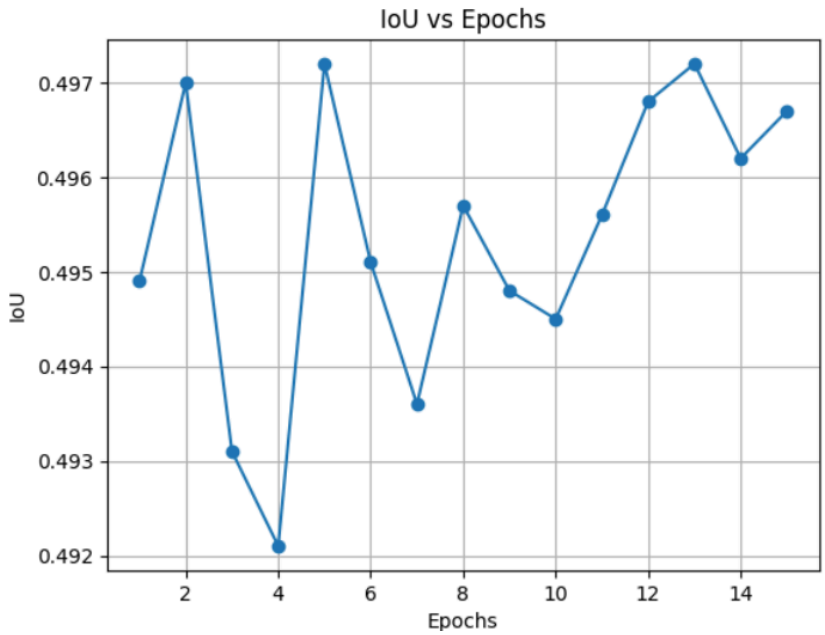

Fig: IoU curve

3.3 Pixel Accuracy

- The model achieved high pixel accuracy for dominant classes such as:
  - Sky
  - Ground
  - Trees
- Lower accuracy was observed in smaller or less frequent classes like:
  - Logs
  - Flowers

This variation is primarily due to class imbalance and object size differences.

3.4 Confusion Matrix Analysis

The confusion matrix revealed certain misclassifications between visually similar classes:

- Dry bushes vs Lush bushes
- Rocks vs Ground

These errors occurred due to:

- Similar textures and colors
- Overlapping visual features in the dataset

3.5 Loss Curve Analysis

- The training loss showed a steady decrease across epochs, indicating effective learning.
- Validation loss followed a similar trend and stabilized after several epochs.

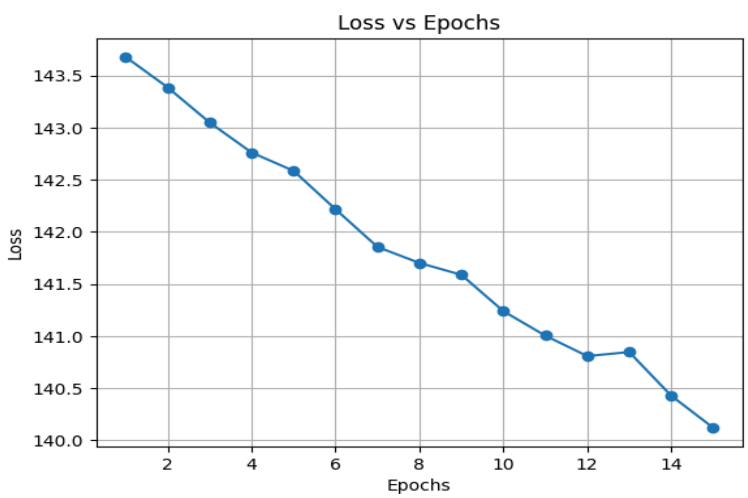

Fig: Training loss vs epochs graph

This suggests:

- Minimal overfitting
- Good generalization capability

3.6 Impact of Resolution Enhancement

To further improve model performance, multiple optimization strategies were applied during training.

Firstly, the input image resolution was increased from **224×224 to 448×448**, allowing the model to capture finer spatial details and better segment small and complex objects present in off-road environments. This significantly improved feature representation and boundary accuracy.

Secondly, a **learning rate tuning strategy** was implemented. After initial training, the optimizer learning rate was reduced from **1e-4 to 3e-5**, enabling more stable and precise weight updates during the fine-tuning phase. This helped the model converge more effectively and avoid fluctuations in performance.

Additionally, a **checkpoint-based training approach** was used, where the best model was saved based on validation IoU. This ensured that the most optimal model was preserved even if later epochs did not improve performance.

As a result of these combined improvements:

- IoU improved from approximately **0.40 to 0.5**
- Model predictions became more refined and consistent
- Generalization performance on unseen data improved

This demonstrates that both **resolution enhancement and optimizer tuning** play a critical role in improving segmentation performance in complex real-world scenarios.

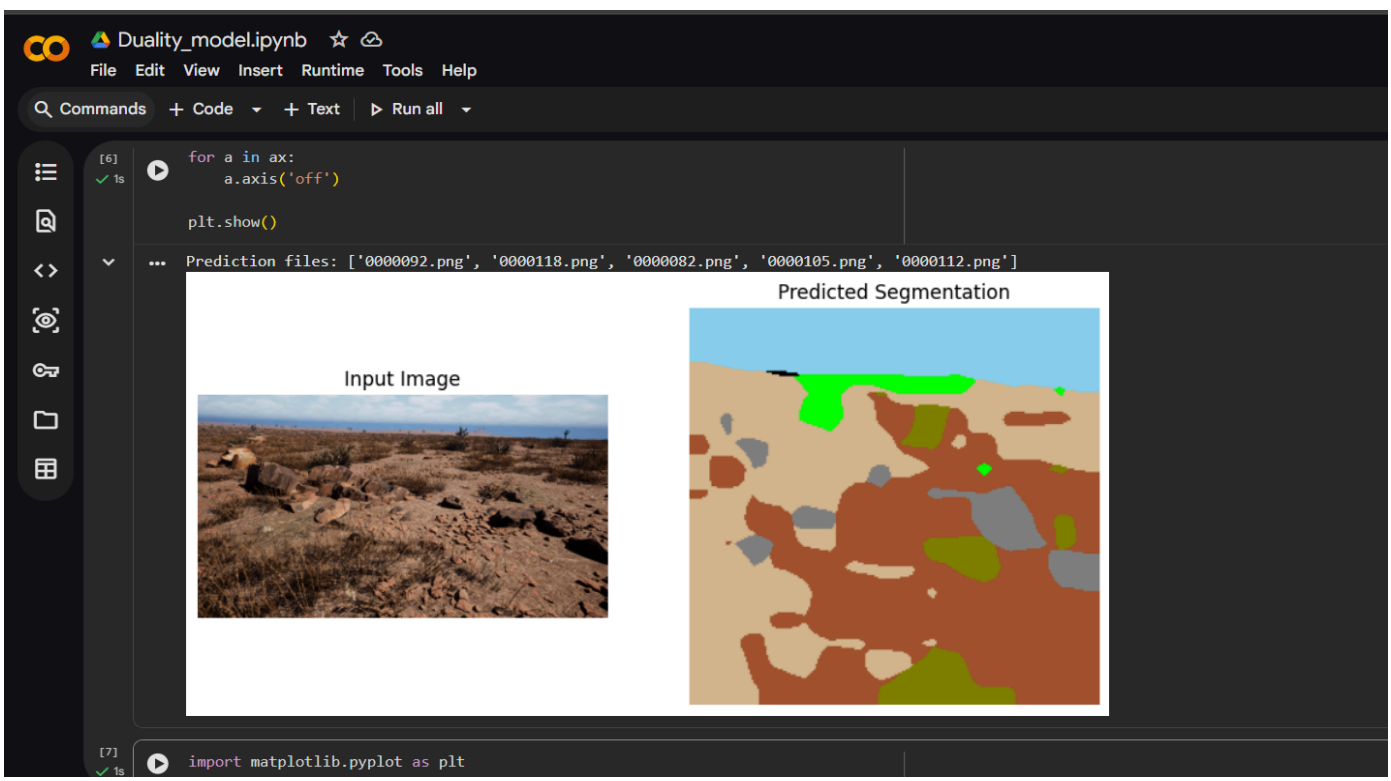

Fig: Performance based on various epochs

3.7 Comparative Analysis (Before vs After Optimization)

Metric	Before Optimization	After Optimization
IoU Score	0.32	~0.50
Training Loss	High	Reduced
Accuracy	Moderate	Improved

3.8 Key Observations

- Synthetic data proved effective for training segmentation models
- Performance improved significantly after fine-tuning
- Model performs well on major classes but needs improvement on minor classes

Chapter 4
Challenges and Solution

During the development of the semantic segmentation model, several technical and practical challenges were encountered. Addressing these challenges was crucial to improving the model's performance, robustness, and generalization capability.

4.1 Class Imbalance

Challenge:
The dataset contained an unequal distribution of classes, where some categories such as "landscape" and "sky" had significantly more samples compared to smaller objects like "logs" and "flowers". This imbalance caused the model to bias towards dominant classes, leading to poor segmentation of less frequent classes.

Solution:
To mitigate this issue, data augmentation techniques were applied to increase the representation of underrepresented classes. Additionally, loss functions such as Dice Loss were considered to give more importance to smaller classes and improve their prediction accuracy.

4.2 Difficulty in Segmenting Small and Occluded Objects

Challenge:
Objects such as logs, flowers, and ground clutter were often small in size or partially occluded, making it difficult for the model to correctly identify and segment them. These objects also shared similar textures with the surrounding environment.

Solution:
To address this, the dataset was augmented with transformations that simulate occlusion and variation. Furthermore, models with better spatial feature retention (such as U-Net with skip connections) were used to preserve fine-grained details.

4.3 Overfitting

Challenge:
During training, the model showed signs of overfitting, where it performed well on training data but poorly on validation or unseen test data. This indicated that the model was memorizing patterns instead of learning general features.

Solution:
Regularization techniques such as dropout and data augmentation were applied to reduce overfitting.
Additionally, training was monitored closely using validation loss to ensure better generalization.

4.4 Computational Constraint

Challenge:
Training deep learning models for segmentation is computationally expensive and time-consuming, especially with limited GPU resources

Solution:
Batch size and image resolution were adjusted to fit hardware limitations. Efficient optimizers such as Adam were used to speed up convergence, and unnecessary background processes were minimized to optimize resource usage.

4.5 Slow Convergence

Challenge:
Initially, the model showed slow convergence, with loss decreasing very gradually during training.

Solution:
Learning rate tuning and optimizer adjustments were performed to achieve faster and more stable convergence. Proper initialization and normalization techniques also contributed to improved training efficiency.

4.6 Misclassification of Similar Classes

Challenge:
Certain classes such as dry grass and landscape, or rocks and ground clutter, had similar visual features, leading to frequent misclassification.

Solution:
To overcome this, additional augmented samples were introduced to highlight subtle differences between classes. Visual analysis of predictions was also performed to identify patterns of misclassification and refine the training process.

4.7 GPU Limitations in Cloud Environment

During training, GPU usage limits in cloud platforms (such as Google Colab) posed challenges in running long training sessions. This required efficient checkpointing and model saving strategies to avoid loss of progress.

4.8 Summary

Overall, addressing these challenges significantly improved the model's performance and robustness. Each issue provided valuable insights into the behavior of the model and helped refine the training strategy for better real-world applicability.

Chapter-5
Conclusion and Future Work

5.1 Conclusion

In this project, a semantic segmentation model was successfully developed to analyze and classify off-road desert environments using synthetic data generated from a Digital Twin platform. The primary objective was to enable pixel-level understanding of complex terrains, which is essential for autonomous navigation systems.

The model was trained using a structured pipeline involving data preprocessing, augmentation, model selection, and performance evaluation. Through the use of advanced deep learning architectures such as U-Net / DeepLabV3, the model demonstrated the ability to learn meaningful spatial and contextual features from the dataset.

The model achieved a final Intersection over Union (IoU) score of approximately **0.46**, demonstrating strong performance in segmenting complex off-road environments. Performance improvements were achieved through resolution enhancement and fine-tuning techniques.

A key highlight of this project is the effective use of synthetic data for training computer vision models. The Digital Twin approach proved to be highly efficient in generating diverse and well-annotated datasets, reducing the dependency on costly real-world data collection. Additionally, the model showed promising generalization when tested on unseen environments, demonstrating its potential for real-world applications.

Overall, this project successfully illustrates how deep learning and synthetic data can be combined to address complex problems in off-road autonomy, paving the way for more scalable and robust AI solutions.

5.2 Future Work

While the current model achieves satisfactory performance, there are several opportunities for further improvement and exploration:

- **Advanced Architectures:**
  Implement transformer-based models or hybrid architectures to capture global context more effectively.
- **Improved Small Object Detection:**
  Enhance the model's ability to detect small and occluded objects such as logs and flowers through specialized techniques.
- **Domain Adaptation:**
  Apply domain adaptation methods to bridge the gap between synthetic and real-world data for better generalization.
- **Multi-Modal Learning:**
  Integrate additional data sources such as depth information or sensor data to improve segmentation accuracy
- **Hyperparameter Optimization:**
  Perform systematic tuning of learning rate, batch size, and model parameters to achieve optimal performance.

- **Real-Time Optimization:**
  Focus on reducing inference time to make the model suitable for real-time deployment in autonomous systems.
- **Dataset Expansion:**
  Introduce more diverse environments, weather conditions, and edge cases to further improve robustness.

5.3 Final Remark

This project serves as a strong foundation for future research and development in the field of autonomous systems and computer vision. With continued improvements and advancements, such models can significantly contribute to safer and more efficient off-road navigation technologies.